Ambulatory ECG Report

Single-lead automated analysis: heartbeats, rhythm, signal quality

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single-lead-ecg-analyzer

PATIENT AND RECORDING

Name / ID	**Example patient**	Recording start	not given (times are from the start)
Age	-	Duration / analyzed	30.1 min / 30.0 min readable
Sex	-	Signal	single lead, 360 Hz
Indication	**Palpitations**	Electrodes	chest (Holter placement)
		Source file	MIT-BIH record 202

KEY MEASURES

Heart rate
71 bpm average
Min 52 bpm at 0:17:00
Max 118 bpm at 0:20:00
1-minute means

Beats analyzed
2,118
Readable time 30.0 min
Unclassified 61

Ventricular extra beats
23 1.1%
46 per hour
Couplets 1 · Runs 0
Bigeminy episodes 0

Supraventricular extra beats
12 0.6%
24 per hour
Couplets 0 · Runs 0
Low reliability (see Methods)

Atrial fibrillation
37.0% burden
Episodes 2
Longest 7.0 min
of the readable time

Pauses of 2 s or longer
0
Longest -
RR interval of 2.0 s or more

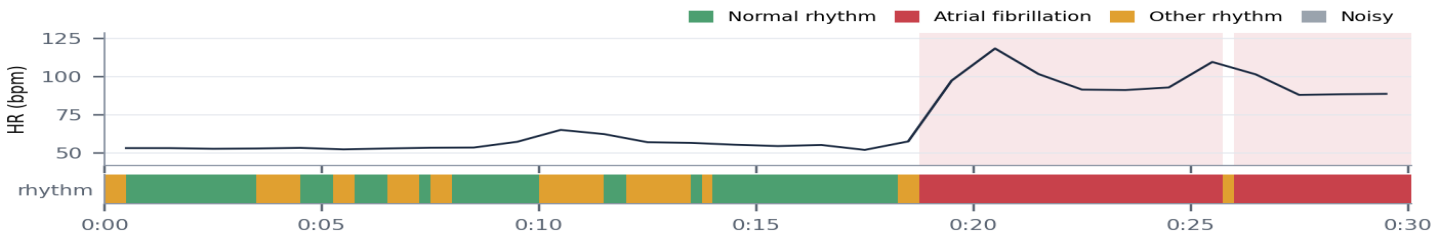
Slow / fast heart rate
0 / 4 min
Minutes below 50 bpm / above 100 bpm
(1-minute means)

Signal quality
99.7% readable
Unreadable 0.1 min
left out of every count

AUTOMATIC FINDINGS

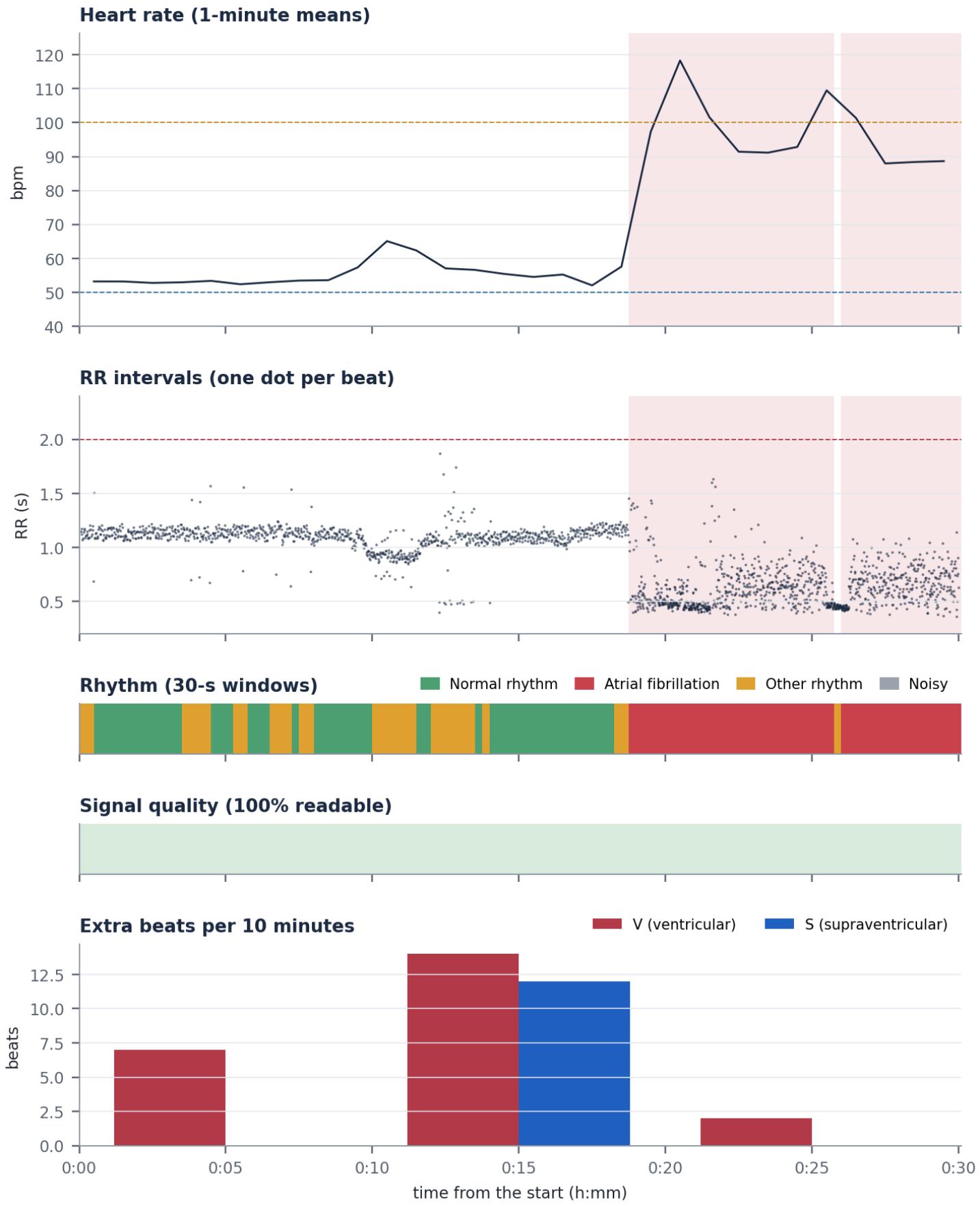
- Atrial fibrillation detected: 37.0% of the readable time, 2 episode(s), longest 7.0 min.
- Ventricular extra beats: 23 (1.1% of beats, 46 per hour), 1 couplet(s).
- Supraventricular extra beats: 12 (0.6% of beats). Low reliability: the beat model detects these poorly (see Methods). Beats during AF are not counted.
- Heart rate: mean 71 bpm, minimum 52 bpm at 0:17:00, maximum 118 bpm at 0:20:00 (1-min means).
- Unclassified beats: 61 (fusion or unclear shape; not counted as extra beats).
- ECG shape screening: not performed. Its models were trained on resting ECGs with limb electrodes and are not valid for chest (Holter) electrode placement.
- Signal quality: the whole recording was readable.

OVERVIEW OF THE WHOLE RECORDING



PHYSICIAN REVIEW AND INTERPRETATION

Reviewed by _____ Signature _____ Date _____



Heart rate: 1-minute means of readable minutes; shaded red = atrial fibrillation episodes. RR intervals: one dot per beat, the line marks the 2-s pause limit. Rhythm: one colour per 30-s window (15-s step). Signal: grey = unreadable 10-s strips, left out of every count.

SUMMARY PER 10-MIN BLOCK

10-min block from	Readable min	HR min	HR mean	HR max	Beats	V	S	AF min	Pauses
0:00:00	10.0	52	53	57	534	7	0	0.0	0
0:10:00	10.0	52	61	97	613	14	12	1.2	0
0:20:00	10.0	88	97	118	971	2	0	9.8	0
Total	**30.0**	**52**	**71**	**118**	**2,118**	**23**	**12**	**11.1**	**0**

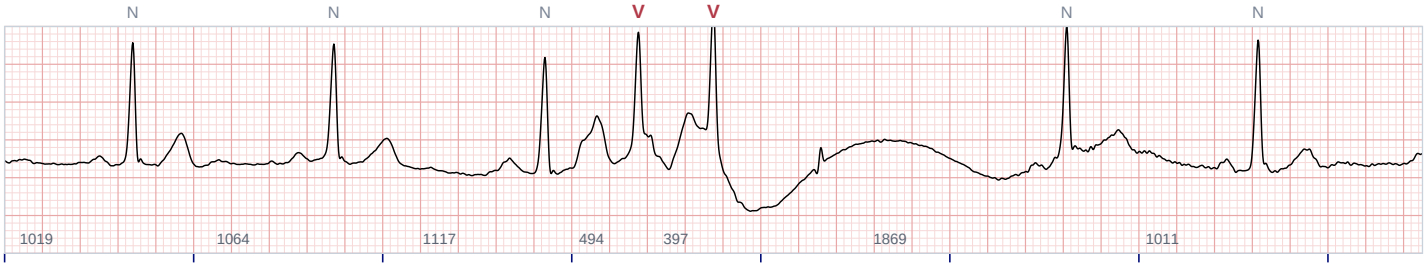
EVENT LOG

Time	Event	Details
0:18:45	Atrial fibrillation	until 0:25:45, 7.0 min
0:20:00	Heart rate above 100 bpm	2 min episode, highest 118 bpm
0:25:00	Heart rate above 100 bpm	2 min episode, highest 110 bpm
0:26:00	Atrial fibrillation	until 0:30:06, 4.1 min

25 mm/s · 10 mm/mV · grid 1 mm / 5 mm · letters: N normal, S supraventricular, V ventricular, U unclassified · numbers: RR interval in ms

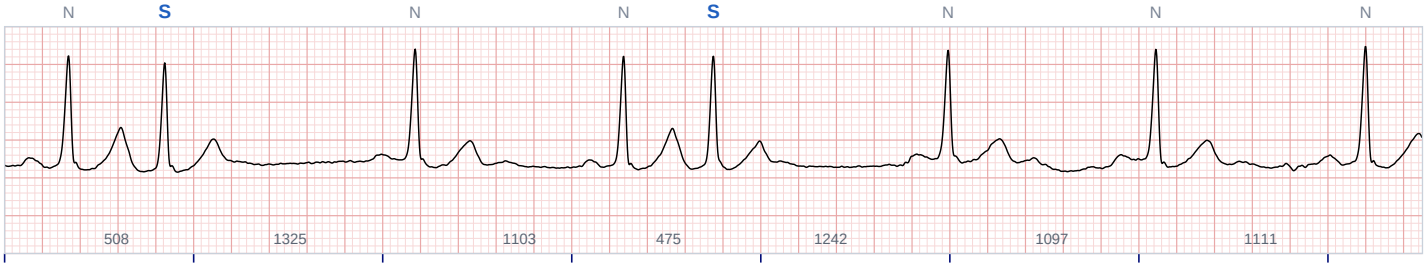
1. Ventricular extra beat (V), most typical example

0:12:13 - 0:12:20 · HR 60 bpm



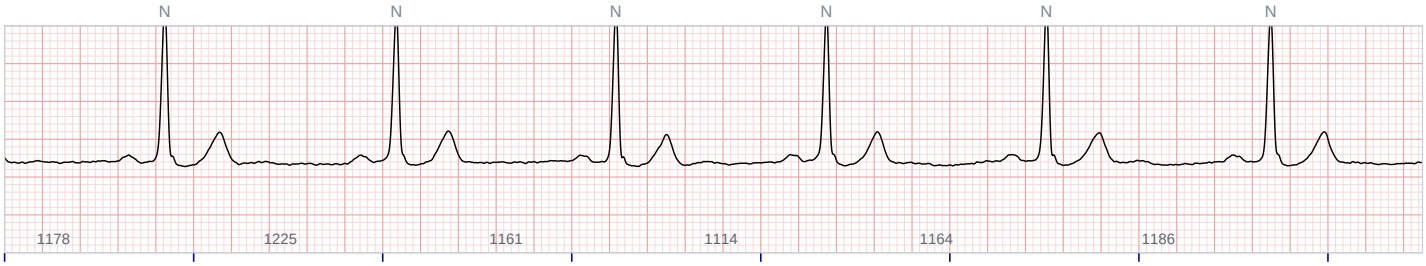
2. Supraventricular extra beat (S), most typical example

0:13:03 - 0:13:10 · HR 61 bpm



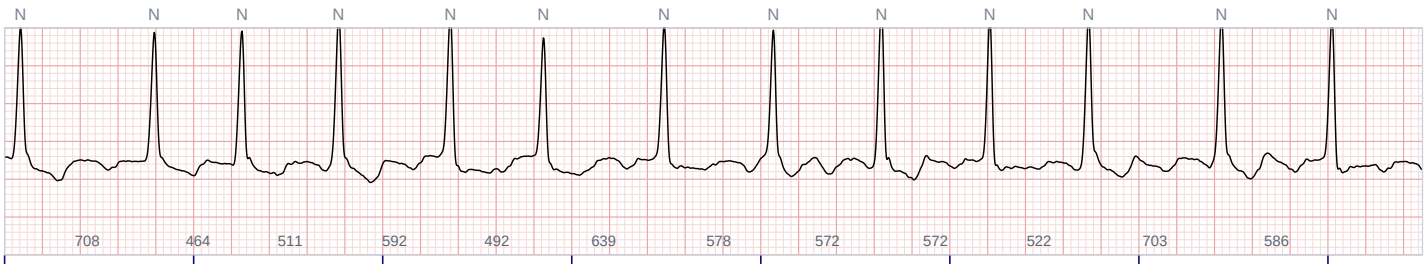
3. Minimum heart rate 52 bpm (1-min mean)

0:17:26 - 0:17:34 · HR 51 bpm



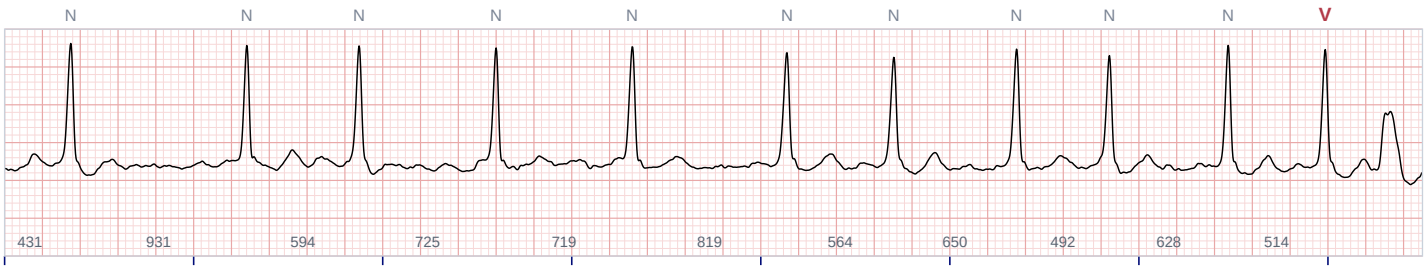
4. Maximum heart rate 118 bpm (1-min mean)

0:20:26 - 0:20:34 · HR 104 bpm



5. Atrial fibrillation, longest episode (7.0 min from 0:18:45)

0:22:11 - 0:22:19 · HR 90 bpm



HOW THE ANALYSIS WORKS

1. Signal quality. The recording is cut into 10-s strips. A quality gate estimates each strip's signal-to-noise ratio; strips below 7.1 dB are marked unreadable and left out of every count, so noise is never reported as a finding.

2. Heartbeats. R peaks are found once (Project 1 detector, 360 Hz). Every beat is classified as N (normal), S (supraventricular extra beat) or V (ventricular extra beat) by a 5-network ensemble that uses the beat shape and the RR intervals around it. Fusion and unclear beats are shown as U.

3. Rhythm. Every 30-s window (15-s step) is classified as normal rhythm, atrial fibrillation (AF), other rhythm or noisy by a 5-network hybrid ensemble (Project 2). AF windows are merged into episodes; the AF burden is AF time divided by readable time. Supraventricular extra beats are not counted during AF.

4. ECG shape (limb electrodes only). Each readable 10-s strip is screened for five diagnostic groups and named findings by single-lead networks trained on resting ECGs (Project 3). Not performed for this recording: on chest (Holter) electrodes these models called almost every strip abnormal, even in a normal-rhythm recording, so their output is not valid there.

5. Measures. Heart rate from RR intervals of consecutive readable beats; 1-minute means need 40 s of readable signal. Pause: RR of 2.0 s or more. Run: 3 or more consecutive extra beats. Bigeminy: an extra beat after every normal beat for 3 or more cycles.

VALIDATION ON PUBLIC DATABASES (PATIENTS NEVER USED IN TRAINING)

Part	Test data	Result
V beats (detector + classifier)	MIT-BIH DS2, 22 patients, 3,095 V beats	Sensitivity 92.8%, PPV 80.3%
S beats	MIT-BIH DS2, 1,824 S beats	Sensitivity 39.6%, PPV 41.2% (low)
Fusion beats (F)	MIT-BIH DS2, 384 F beats	Sensitivity 0%: not detected
Normal beats	MIT-BIH DS2, 44,195 N beats	Sensitivity 95.4%, PPV 96.9%
Atrial fibrillation	CinC 2017 test, 152 AF recordings	Sensitivity 84.2%, PPV 79.5%, specificity 97.9%
	MIT-BIH DS2, 2,640 30-s windows	Sensitivity 93.9%, PPV 75.2%, specificity 94.8%
	MIT-BIH AF DB, 56,207 30-s windows	Sensitivity 93.7%, PPV 97.2%, specificity 98.2%
ECG shape, 5 groups	PTB-XL test (internal), lead II	Mean AUC 0.864
(limb electrodes)	Chapman-Shaoxing (external), lead II	Mean AUC 0.852
Quality gate	PTB-XL test + real electrode noise	Catches 94% of strips below 0 dB; 6% false rejections

LIMITATIONS

- A single lead cannot replace a 12-lead ECG or a cardiologist's review; findings are prompts for review, not diagnoses.
- S beats are detected poorly (sensitivity 40%) and fusion beats not at all; 4.6% of normal beats get another label.
- Frequent extra beats can be mistaken for AF: MIT-BIH records 200 (27% false AF burden), 228 (19% false AF burden), 232 (17% false AF burden) in the external test. The report warns when AF episodes contain many ventricular extra beats.
- The ECG-shape screening was validated only on resting 10-s ECGs with limb electrodes, and its operating points have modest PPV (for example MI 50%, hypertrophy 26%).
- Validated on public databases only. Not yet tested on the AD8232 hardware or on new patients.

SOFTWARE

Engines (GitHub, pinned commits): Project 1: ecg-arrhythmia-detector (8909557); Project 2: ecg-af-detector (4554887); Project 3: ecg-lead2-diagnosis (246bed3). Analysis time 13 s on CPU.